HOSTEL MANAGEMENT SYSTEM (MERN) - DEVELOPMENT PLAN

1. JWT Authentication

User/Admin login with token-based authentication. Secure routes using middleware.

2. Email OTP System

Used for registration verification and password reset. OTP sent via email and validated before access.

3. Role-Based Access Control

Two roles: Admin and User. Admin can manage rooms, users, complaints. Users can book rooms and raise complaints.

4. Dashboard

User Dashboard: Profile, bookings, complaints. Admin Dashboard: total users, rooms, complaints, revenue.

5. Room Availability Filter

Filter rooms based on availability, capacity, and fees. Only show rooms with free seats.

6. Booking Logic

One user can book only one room. Room occupancy should not exceed capacity.

7. Complaint System

Users can raise complaints. Admin updates status: Pending, In Progress, Resolved.

8. Feedback System

Users can submit feedback with rating. Admin can view feedback.

9. Payment & PDF Receipt

Store payment details and generate PDF receipt using PDFKit.

10. Suggested Timeline (4 Weeks)

Week 1: Setup project, authentication, database Week 2: Room, booking, dashboard Week 3: Complaint, feedback, role access Week 4: Email OTP, PDF, testing